

4.6 Normal Distribution

A frequently occurring continuous random variable is the random variable X , which follows the normal distribution, i.e. is defined using the following probability density function:

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad \text{for } -\infty < x < \infty$$

Example 5

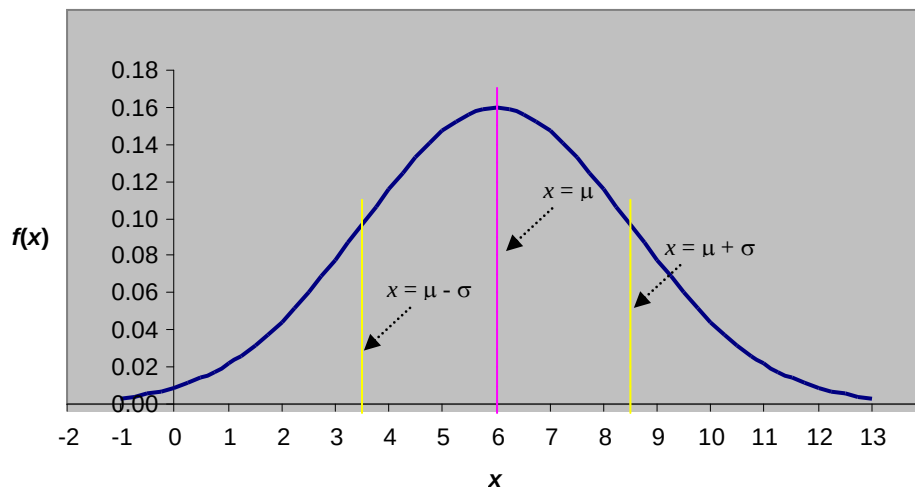


Figure 4.5 Normal probability density function with $\mu = 6$ and $\sigma = 2.5$

Remarks

- The random variable X is a normal random variable with parameters μ and σ .
- Notation: $X \sim N(\mu; \sigma^2)$
- $f(x)$ has a **maximum value** at $x = \mu$.
The vertical line $x = \mu$ is an **axis of symmetry** of $f(x)$.
At $x = \mu + \sigma$ and at $x = \mu - \sigma$, the graph of $f(x)$ shows a “kink”; these points are the so-called **points of inflexion**.
In other words, the number μ is a measure for central tendency and the number σ is a measure for variability.
- Using standard integrals, the following results can be obtained:

$$\begin{aligned} E(X) &= \mu \\ V(X) &= \sigma^2 \end{aligned}$$

- The special case $\mu = 0$ and $\sigma = 1$ is called the standard normal distribution. In that case the variable X is denoted by Z . The cumulative distribution function is denoted as $\Phi(z)$.

So, if $Z \sim N(0; 1)$, then $\Phi(z) = P(Z \leq z)$. E.g. $\Phi(0.5)$:

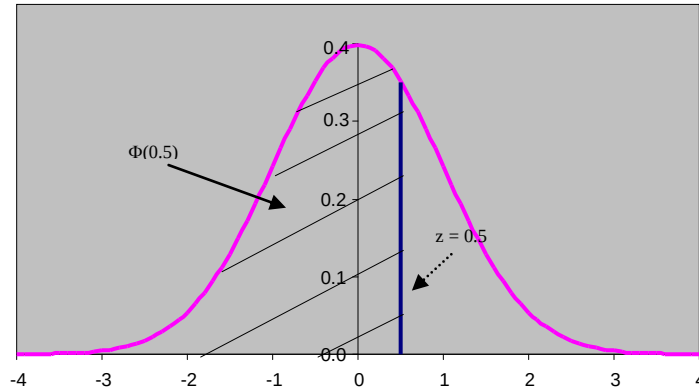


Figure 4.6 Probability that $Z \leq 0.5$ for a standard normal random variable

- $\Phi(z)$ is determined by using **Table III** or by using a **graphical calculator**.

Table III:
gc/distr/option 2: $\Phi(z)$ for $-3.99 \leq z \leq 3.99$
 $\Phi(z) = \text{normalcdf}(\text{lower bound}, \text{upper bound}, \mu, \sigma)$
 with: lower bound = $-1E99$ or -10^{10} or $-\infty$
 upper bound = z , $\mu = 0$ and $\sigma = 1$

Example 6

- $P(Z \leq 2.58) = \Phi(2.58) = 0.995060 \approx 0.9951$ **Table III**
- $P(-0.52 \leq Z \leq 1.11) = \Phi(1.11) - \Phi(-0.52) = 0.866500 - 0.301532 \approx 0.5650$ **Table III**
- $P(-0.52 \leq Z \leq 1.11) = \text{normalcdf}(-.52, 1.11, 0, 1) = 0.5650$ **gc/distr/2**
- $P(Z \geq -1.168) = P(Z \leq 1.168) = 0.8786$ **symmetry Table III; interpolation**
- $P(Z \geq -1.168) = \text{normalcdf}(-1.168, 10^{10}, 0, 1) = 0.8786$ **gc/distr/2**
- $P(-a \leq Z \leq a) = 0.9 \Leftrightarrow a = \text{invNorm}(.95, 0, 1) = 1.6449$ **gc/distr/3**
- $P(-a \leq Z \leq a) = 0.9 \Leftrightarrow a = 1.645$ **Table III; interpolation**

Arbitrary normal random variables

If the normal random variable X is **not** a standard normal random variable (i.e. $\mu \neq 0$ or $\sigma \neq 1$), then the following theorem will be used

Theorem

- If $X \sim N(\mu; \sigma^2)$, then $Z = \frac{X-\mu}{\sigma} \sim N(0; 1)$
- If $X \sim N(\mu; \sigma^2)$, then $F(x) = \Phi\left(\frac{x-\mu}{\sigma}\right)$

Example 7

Suppose body length = $X \sim N(\mu = 170; \sigma^2 = 225)$. Determine $P(X \geq 160)$.

Solution

Using Table III:	$P(X \geq 160)$	=	$P(Z \geq \frac{160-170}{15})$	
		=	$P(Z \geq -0.6667)$	
		=	$P(Z \leq 0.6667)$	symmetry
		=	0.7475	interpolation
Using gc/distr/2:	$P(X \geq 160)$	=	<code>normalcdf(160, 10^10, 170, 15)</code>	
		=	0.7475	

Intervals of the form $\mu \pm k \cdot \sigma$

Let $X \sim N(\mu; \sigma^2)$.

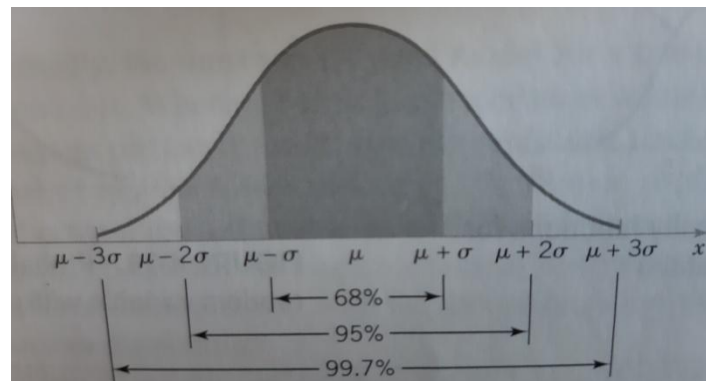


Figure 4.7

Probabilities associated with a normal distribution

$$\begin{aligned} P(\mu - \sigma < X < \mu + \sigma) &= P\left(\frac{(\mu - \sigma) - \mu}{\sigma} < \frac{(X - \mu)}{\sigma} < \frac{(\mu + \sigma) - \mu}{\sigma}\right) \\ &= P(-1 < Z < 1) \\ &= \text{normalcdf}(-1, 1, 0, 1) = 0.6827 \end{aligned}$$

In a similar way one can find expressions for the intervals $\mu \pm 2 \cdot \sigma$ en $\mu \pm 3 \cdot \sigma$.

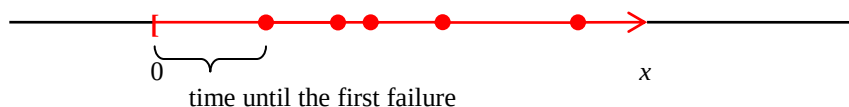
Summarizing

$$\begin{aligned} P(\mu - \sigma < X < \mu + \sigma) &= 0.6827 \\ P(\mu - 2\sigma < X < \mu + 2\sigma) &= 0.9545 \\ P(\mu - 3\sigma < X < \mu + 3\sigma) &= 0.9973 \end{aligned}$$

It is therefore almost impossible that a measurement x is **more than 3σ** away from μ .

4.8 Exponential Distribution

Given is a Poisson process with parameter λ = the average number of failures per time unit.



Let the **continuous** random variable X denote the time until the first failure. Then

$$\begin{aligned} P(X > x) &= P(\text{zero failures during } [0 ; x]) \\ &= P(N = 0) \\ &= e^{-\lambda x} \cdot \frac{(\lambda x)^0}{0!} = e^{-\lambda x} \end{aligned} \quad N \sim \text{Poi}(\lambda x)$$

So

$$F(x) = P(X \leq x) = 1 - e^{-\lambda x}$$

and

$$f(x) = \frac{d}{dx} F(x) = \lambda \cdot e^{-\lambda x} \quad \text{for } x > 0$$

The random variable X is an exponential random variable with parameter λ .

Example 8

Given is an exponential random variable X with parameter $\lambda = 0.1$. Determine $P(X > 5)$.

Solution

$$\begin{aligned} P(X > 5) &= \int_5^{\infty} 0.1 \cdot e^{-0.1x} dx \\ &= [-e^{-0.1x}]_{x=5}^{x=\infty} \\ &= 0 - (-e^{-0.5}) = e^{-0.5} \approx 0.6065 \end{aligned}$$

Alternative methods

- $$P(X > 5) = \int_5^{100} 0.1 e^{-0.1x} dx = 0.6065 \quad \text{gc/math/9}$$

NB 100 instead of ∞
- $$\begin{aligned} P(X > 5) &= 1 - F(5) \\ &= 1 - (1 - e^{-0.1 \cdot 5}) = e^{-0.5} \end{aligned}$$

Remarks

- Notation: $X \sim \exp(\lambda)$
- Using **integration by parts** one can find expressions for the **expected value** and the **variance** of an exponential random variable X .

$$E(X) = \frac{1}{\lambda} \quad \text{See video presentation 4-37}$$

$$V(X) = \frac{1}{\lambda^2} \quad \text{Idem}$$

- The **lack of memory property** of an exponential distribution:

$$\begin{aligned} P(X < x + a \mid X > x) &= \frac{P(x < X < x + a)}{P(X > x)} \\ &= \frac{(1 - e^{-\lambda(x+a)}) - (1 - e^{-\lambda x})}{e^{-\lambda x}} \\ &= 1 - e^{-\lambda a} = P(X < a) \end{aligned}$$

So after waiting for x minutes without a failure, the probability of a failure in the next a minutes is the same as the probability of a failure in the a minutes immediately after starting the counter. Therefore $X =$ **length of the failure-free period**.

Example 9

Customers are entering a department store with a frequency of 120 per hour. Determine the probability that there will be more than 90 seconds between two consecutive customers.

Solution

So $P(X > 1.5 \text{ minutes}) = ?$, for $X \sim \exp(\lambda = 2 \text{ per minute})$

$$\begin{aligned} \text{Thus } P(X > 1.5) &= 1 - F(1.5) \\ &= 1 - (1 - e^{-2 \cdot 1.5}) \\ &= e^{-3} \\ &= 0.0498 \end{aligned}$$